

Kumpel et al. (2024)

The diagram illustrates the metabolic pathways of sulfur in the periplasm, membrane, and cytoplasm. Key components and reactions include:

- Periplasm:**
 - Sox Pathway:** SoxB and SoxYZ form a complex. SoxB oxidizes H_2O to SO_4^{2-} and 2H^+ . SoxYZ is reduced from S^0 to S^{2-} . SoxAX and SoxCD are also involved in the cycle, with SoxCD being re-oxidized by SoxYZ.
 - SoxYZ-S-SO₃⁻ Pathway:** SoxYZ-S-SO₃⁻ is re-oxidized to SoxYZ-S-SO₃⁻ by SoxAX, releasing 2e^- . SoxYZ-S-SO₃⁻ is then re-oxidized to SoxYZ-S-SO₃⁻ by SoxB, releasing 2H^+ and SO_4^{2-} .
 - SoxYZ-S-SO₃⁻ Pathway:** SoxYZ-S-SO₃⁻ is re-oxidized to SoxYZ-S-SO₃⁻ by SoxB, releasing 2H^+ and SO_4^{2-} .
- Membrane:**
 - SoeABC:** MQ → MQH₂
 - DoxDA:** MQ → MQH₂
- Cytoplasm:**
 - Dsr:** R-S-S- → HSO₃⁻ → SO₄²⁻
 - sHdr:** sHdrB1, sHdrB2, sHdrC1, sHdrC2, sHdrA, sHdrH (type I); sHdrB1, sHdrB3, sHdrC1, sHdrA, sHdrH (type II)
 - TsdA:** 2S₂O₃²⁻ → S₄O₆²⁻
 - SorAB/SorT:** SO₃²⁻ → SO₄²⁻